

Is Holding Better? A Gesture-Elicitation Study with Controllers in Virtual Reality

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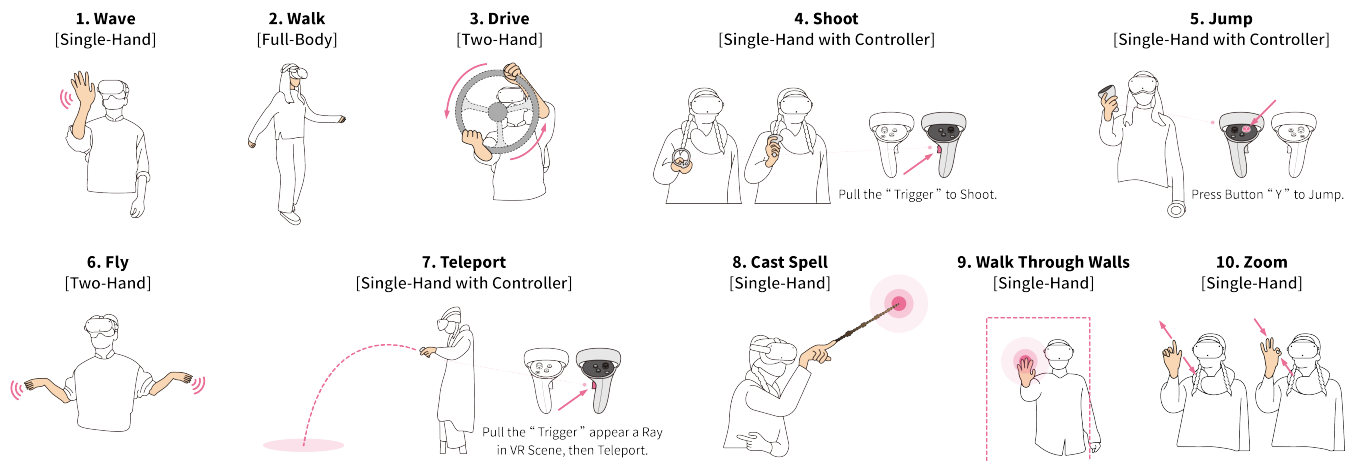


Figure 1: This study's main contribution is exploring a Gesture Elicitation Studies method incorporating VR controllers. Via agreement score (AR) calculation, a virtual reality movement consensus gesture set with 10 tasks is proposed.

Abstract

This study is the first to integrate VR controllers into the originally hand-based Gesture Elicitation Studies (GES), exploring the protocols and preferred gestures for performing virtual reality movement using different body parts under barehanded and controller-based conditions. We also calculated protocol scores and attempted to understand the reasons behind participants' preference choices. Contrary to the expectation that participants would all prefer barehanded gestures, the study revealed that while most participants preferred barehanded gestures, they preferred the single-handed

controller for tasks such as shooting, jumping, and teleporting due to haptic feedback, multifunctionality, precision, and ease of use. The study also presented a consensus gesture set that includes controller-based gestures, filling the gap in virtual reality gesture elicitation study regarding the combination of controllers and hand gestures.

CCS Concepts

• Human-centered computing → Human computer interaction (HCI); • HCI design and evaluation methods → Usability testing.

Keywords

Gesture Elicitation Studies (GES), Virtual Reality Controllers, Barehanded Interaction, User Preferences in VR Gestures, VR Interaction Design

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1 Introduction

In recent years, the two most common methods of interacting with virtual reality (VR) are controller-based interaction and gesture-based interaction. The first involves using VR controllers combined with users' movements, the most mainstream and relatively well-developed interaction method. The second method, directly manipulating the VR environment through hand gestures or body gestures, is more direct and natural, providing a more intuitive and immersive experience [7]. Some research has investigated the most widely chosen gestures for various commonly used VR operations using "Gesture Elicitation Studies (GES)" [6, 22, 23]. GES, initially proposed and implemented by Wobbrock et al., asks participants to design gestures corresponding to specific referents (the effects of gestures) [21]. The value of GES lies in its ability to provide researchers with deep insights into users' capabilities and preferences. It permits participants to interact with the system in their chosen manner, directly involving end-users in the design process [20]. However, GES in VR design have only focused on bare hands scenarios, implicitly assuming that users prefer bare hands solutions over controller-based ones in VR. Some researchers have compared bare hands gestures with controller use. However, the researchers defined the gestures, which may have overlooked the most widely chosen gestures for controller manipulation, affecting the comparative results. Currently, there is a lack of research that simultaneously examines the optimal gesture design methods for "barehanded-based" and "combining VR controllers-based" gesture interactions.

Accordingly, investigations are needed to explore comparing the most widely chosen gestures and user preferences. To address this issue, we incorporated comparisons with handheld controllers into previous bare hands gesture elicitation studies while involving participants in gesture elicitation using different body parts (single-hand, two-hands, full-body). We compared the gestures created under these conditions, focusing on two aspects: participants' preferences for the manipulation methods under the two conditions and the reasons for the differences in preferences. In summary, this article presents the following findings and contributions:

- (1) For the first time, controllers were incorporated into Gesture Elicitation Studies (GES), exploring a broader range of design gestures.
- (2) A consensus gesture set for 10 tasks was generated by performing each task using both bare-hand and controller-based GES.
- (3) For complex or effort-intensive movements, controllers gained participant preference by enabling multifunctionality and alternative movements through button use.

2 Related Works

This section will elaborate on the existing research on Gesture Elicitation Studies (GES) in virtual reality, as well as the studies comparing bare-hand and controller-based interactions.

2.1 Research on Gesture Elicitation Studies in Virtual Reality

In the past few years, Gesture Elicitation Studies (GES) have been applied to virtual reality (VR). Some studies have explored interactions for selection and manipulation [2, 11, 16]. Wu et al. investigated task gestures used in virtual shopping scenarios [23]. Arora discussed gestures for controlling animation effects in animation production [1], while Ganapathi considered movement gestures in virtual reality under seated conditions [6]. Additionally, GES in VR has focused on designing gestures using different body parts. Most studies involve single-hand or two-hand (upper-body) gestures [6], while a small number of studies address full-body gestures [16]. The involvement of different body parts influences users' sense of immersion and the physical effort required for the movements. This will also impact participants' choices between gesture-based and controller-based forms when designing movements. Therefore, subsequent experiments also consider the factor of body part involvement.

Currently, All gestures designed in GES for VR have been designed to be completed with bare hands. This might be due to the influence of previous research conducted on other devices. However, the VR environment differs from traditional interaction environments. As the default interactive device, VR controllers offer button and grip functions. When considering gesture design, we should not overlook the benefits brought by VR controllers but instead integrate them into gesture-elicitation design.

2.2 Comparison of VR Control Methods

With the introduction of gesture control in VR systems, comparative studies between hand tracking and traditional controllers in VR tasks have increased, revealing their pros and cons in terms of naturalness [17], accuracy, efficiency [13], and user experience [15]. Overall, hand tracking is outstanding in improving immersion and realism through natural and intuitive interactions [12], whereas traditional controllers dominate in terms of precision, efficiency, and stability. Barehanded interaction is generally considered more natural and enhances users' sense of immersion and presence. Hameed found that hand tracking is more intuitive in "reach-pick-place" tasks due to its simulation of natural real-world movements [7]. Furthermore, Pangestu et al. discovered that hand tracking significantly enhances the realism of interaction in Gamelan (Saron) application studies [17]. Neamoniti and Kasapakis's research suggests that although hand tracking is slightly inferior to movement controllers in terms of learnability and usability, it performs comparably in immersive gaming experiences (e.g., GEQ core module) and task load (e.g., NASA-TLX) [15]. Luong et al. also noted that users prefer hand tracking in touch tasks, finding it more intuitive [12].

In contrast, traditional controllers perform better on most performance metrics. Masurovsky et al. found that in "grab-place" tasks, traditional controllers significantly outperform hand tracking regarding accuracy, speed, error rate, and subjective measures like comfort, precision, and satisfaction [13]. Hameed et al. also discovered that controllers reduce users' mental load and improve task efficiency [7]. Moreover, Johnson confirmed that controllers outperform body tracking (including hand and eye tracking) in terms

of accuracy, efficiency, and user satisfaction for button-selection tasks [10].

Although these studies provide detailed comparisons of hand tracking and controllers regarding performance, task outcomes, and user satisfaction, they don't consider the optimal bare hands and controller gestures. In other words, the above discussions are based on expert-designed gestures rather than the natural choices users would intuitively make. Additionally, existing studies mainly focus on the performance or satisfaction of the two interaction methods without delving into users' underlying preferences and reasons for their choices based on the characteristics of each method.

3 Experimental Design

This section will briefly describe the design and implementation of gesture elicitation research in our experiment from four aspects: referent, participants, instruments and equipment, and experimental process.

3.1 Referent

We collected movements from the Steam and Oculus platforms, relevant VR works from the 2024 SIGGRAPH, the 2024 Taiwan Innotech Expo, and related literature on movement design [9, 22]. Through a discussion by a four-person expert focus group, we selected 10 commonly observed task movements as the referent for this study (Table. 1). The referent included daily movements, non-daily movements, and movements impossible in real life. During the experiment, researchers informed participants of the required movement names orally.

Table 1: 10 commonly observed gestures as the referent for this study

1. Wave	6. Fly
2. Walk	7. Teleport
3. Drive	8. Cast Spell
4. Shoot	9. Walk Through Walls
5. Jump	10. Zoom

3.2 Participants

Based on the literature review by Cheng [5], most Gesture Elicitation studies involve approximately 20 participants. However, due to the challenges of recruiting participants in VR environments and the exploratory nature of this preliminary study, we recruited 10 participants (3 male and 7 female). All participants were university students aged between 20 and 30 with a design background and had basic or advanced VR usage experience.

3.3 Instruments and Equipment

The VR device used in this experiment was the Meta Quest 2, offering a resolution of 1832 x 1920 for both eyes. The device was set to the VR home page and was responsible for providing a VR environment where participants could design VR gestures in an immersive setting (Fig. 2b). Furthermore, a camera captured video recordings of participants' gestures (Fig. 2a).



Figure 2: (a) Experiment Setup, and (b) the VR environment in which the participants are located while designing VR poses.

3.4 Experimental Process

The experiment used a within-subject design to avoid order effects, with participants randomly assigned to two groups: Group A first completed hand-only gestures followed by VR controller gestures, while Group B did the opposite. At the start of the experiment, participants were briefed on the procedure, tasks, and requirements and gave informed consent. Researchers assisted participants in wearing the VR equipment and instructed them to design a single-hand, a two-hand, and a full-body gesture for each task. The "think-aloud" method was used to gather design principles, and participants were asked to explain the reasons for selecting specific gestures. After completing five movement designs, participants removed the headsets to rest and ranked the effort required and preferences for the designed gestures. The effort was assessed using an adapted version of the RPE scale [3]. A semi-structured interview followed, covering questions about 1. preferences, 2. the least effortful movements, 3. comparisons between VR controller and hand-only movements, and 4. the differences between single-hand, two-hand, and full-body movements. The above steps were repeated until all movements were completed.

4 Result

4.1 Data Analysis and Agreement Scores (AR)

From 10 participants performing 10 VR tasks, we collected 600 gestures (10*10*6), divided equally into 100 gestures for a single-hand, the single-hand with controllers, two-hands, two-hands with controllers, full-body, and full-body with controllers. Subsequently, three researchers conducted a brainstorming session and performed a two-step classification of a user-defined set of gestures: First, gestures with the same shape and/or trajectory were grouped into a single gesture. Second, for gestures that had similar characteristics in shape and/or trajectory but were controversial, the three experts replayed the corresponding video files collected during the design process and referred to the participants' verbal explanations to vote on them. If two or more experts agreed, the gestures were considered to belong to the same group. Using Wu's [23] approach, we calculated agreement scores (AR) (Fig. 3) for each target task based on the updated consensus formula by Vatavu and Wobbrock [18]. (Due to significant differences among the types of gestures, we calculated agreement scores only within the same category.)

$$AR(r) = \frac{|P|}{|P|-1} \sum_{P_i \in P} \left(\frac{|P_i|}{|P|} \right)^2 - \frac{1}{|P|-1}$$

The average AR across 10 VR tasks was 0.116±0.157 (Fig. 4b). Among them, gesture AR with controllers was 0.056±0.071, while those without controllers were 0.177±0.193, indicating that bare-handed gesture design is more likely to achieve unified gestures. Divided by body parts, single-hand gestures had an AR of 0.129±0.139, two-hand gestures scored 0.122±0.178, and full-body gestures scored 0.097±0.156. Comparatively, single-hand and two-hand gestures tend to result in more consistent designs. Looking at the highest AR across different tasks (Fig. 4a), two-hand and full-body walking (0.622) and two-hand driving (0.622) scored above 0.5, which, according to Vatavu and Wobbrock's interpretation of AR magnitudes [24], falls under "very high agreement." Single-hand shooting (0.322) and single-hand zooming (0.356) scored between 0.3-0.5, which is categorized as "high agreement". Two-hands waving (0.200) and full-body jumping (0.233) were in the 0.1-0.3 range, signifying "medium agreement". Two-hands and full-body flying (0.067), single-hand controller teleportation (0.044), single-hand casting spell (0.067), and two-hands walk through walls (0.089) scored below 0.1, classified as "low agreement."

4.2 Effort Level of Gesture

A paired-sample T-test shows that the average effort level for bare-hand gestures is 3.51, while for gestures with VR controllers, the average is approximately 3.63. The effort level for the controller group is slightly higher than bare-hand, but the difference is not significant (t=-1.02, p=0.33).

4.3 Most Choices and User Preference for Interaction

Considering the differences in modes and body parts, each task ultimately produced six the most widely chosen gestures. Meanwhile, we selected the gesture type most favored by participants and combined both to provide the best-recommended gesture for

Task	Body Part	Single-hand	Two-hands	Full Body
	Design By			
1. Wave	Bare-hand	0.156	0.200	0.044
	Controller-based	0.000	0.022	0.000
2. Walk	Bare-hand	0.333	0.622	0.622
	Controller-based	0.111	0.089	0.244
3. Drive	Bare-hand	0.456	0.622	0.000
	Controller-based	0.022	0.067	0.000
4. Shoot	Bare-hand	0.322	0.089	0.089
	Controller-based	0.222	0.089	0.022
5. Jump	Bare-hand	0.089	0.089	0.233
	Controller-based	0.211	0.022	0.133
6. Fly	Bare-hand	0.044	0.067	0.067
	Controller-based	0.022	0.022	0.022
7. Teleport	Bare-hand	0.000	0.022	0.000
	Controller-based	0.044	0.000	0.000
8. Cast Spell	Bare-hand	0.067	0.022	0.067
	Controller-based	0.022	0.022	0.000
9. Walk Through Walls	Bare-hand	0.044	0.089	0.044
	Controller-based	0.067	0.022	0.000
10. Zoom	Bare-hand	0.356	0.111	0.333
	Controller-based	0.000	0.156	0.022

Figure 3: Agreement Scores (AR) for Each target task.

each task. As shown in the Figure 1 shows that single-hand gestures are the best for waving, spell casting, walking through walls, and zooming. The single-hand with controller gestures are the best for shooting, jumping, and teleporting. Two-hand gestures are optimal for driving and flying, while full-body gestures are the best for walking. Meanwhile, based on user preference rankings, the most disliked gesture design involves full-body gestures combined with VR controllers.

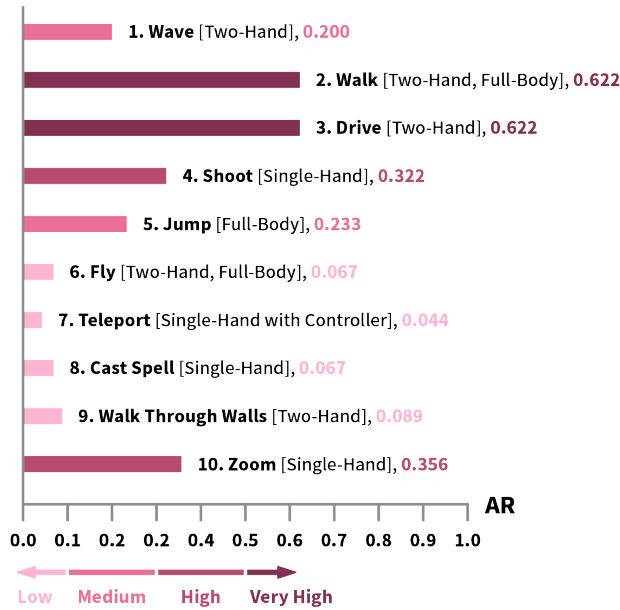
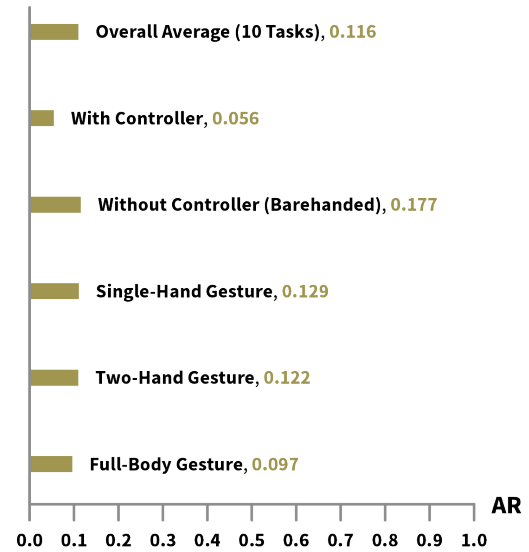
A. Task(s): Category Agreement Score (AR)**B. Task(s): Task Agreement Score (AR)**

Figure 4: (a) the highest AR across different tasks, (b) The average AR across 10 VR tasks.

5 Discussion

5.1 Reasons for Gesture Agreement Score Differences

Daily movements in real life that can be seamlessly implemented in virtual reality are intuitively used by participants to complete gesture design. Consequently, full-body walking and two-handed driving obtained the highest agreement scores. In single-handed shooting, some participants designed gestures with the hand symbolizing a gun, while others simulated the gesture of holding a gun. Single-handed zooming, influenced by technical legacy issues [14], led participants to draw on zooming methods from tablets, mobile devices, or movement-sensing games. Yet, the agreement scores for both movements exceeded 0.3.

Movements that cannot occur in real life are expressed in various ways in virtual reality. Agreement scores for two-hand and full-body flight, single-hand with controller teleportation, single-hand spellcasting, and two-hand walking through wall were all below 0.1. In the gesture design, we referred to the most common form of referents in [5], which were provided in textual form. This gave participants a large space for imagination, leading to the creation of a variety of gestures. For example, flying gestures were inspired by Superman, birds, immortals, paragliders, airplanes, and remote control interfaces; spell-casting gestures included hands symbolizing magic wands and gestures representing attacks, fireballs, and transformations. The diversified gestures led to extremely low agreement scores in the end. To improve the consistency of gestures for such movements, it might be necessary to adopt a framework-based design approach to establish relevant contexts [22] or to present

the outcomes before gesture design [1], fostering consensus among participants and leading to convergent gestures.

5.2 Reasons for Differences in Preferred Movement Category

The single and bare-hand mode meets the demands of “effort-saving,” “intuitiveness,” and “quick activation,” and most participants preferred using “single and bare-hand” gestures for swinging, zooming, spell-casting, and phasing through walls. The gestures for waving and zooming are consistent with the consensus gesture set summarized by Hosseini [8]. P5 emphasized, “Zooming with one hand is sufficient; using two-hand is unnecessary and redundant,” which aligns with their everyday habits (e.g., pinch-to-zoom on smartphones, simple gesture controls) and avoids additional physical or learning costs.

Contrary to the traditional perspective in Gesture Elicitation Studies [8], which favors bare hands, we discovered that the single-handed controller mode was preferred when it fulfilled the requirements of “haptic feedback,” “multifunctionality,” “high-precision tasks,” and “effort-saving” making it the preferred choice for most participants in tasks like shooting, jumping, and teleporting. Firstly, gripping a controller equipped with tactile texture or vibration functions fulfills users’ expectations for “real weapon operation,” adding to their sense of immersion and safety [19]. P6 noted, “Using a controller feels like holding a gun.” Controllers also provide more diverse functionalities and higher operational precision [13]. P8 stated, “Controllers have buttons or sensors that allow for more precise control of direction, force, aiming, and firing.” This demonstrates that controllers can simulate complex manipulation processes through buttons and sensors, which are difficult to achieve

with single-handed bare-hand gestures. Finally, single-handed controllers simplify movements such as “jumping” and “teleporting,” which do not exist in reality or are physically demanding, by offering a one-button solution compared to single-hand gestures in video games [9].

Two-hand or full-body movements, implemented in continuous gestures [8], offer greater embodiment and immersion. In studies of action gestures in Kinect video games, when involving lower body movements or actions that involve leaving the ground (such as walking or flying), participants tend to use one-handed gestures to complete them [9]. In this study, the VR environment allows more body parts to participate, with participants preferring to use both hands for flying and driving and full-body movements for walking. During flying, as P9 mentioned (when both hands are spread out, the sense of immersion in flight is immediately felt), in first-person dominated VR activities, participants prefer more embodied forms of interaction [4]. Moreover, they expressed concerns that although the “single-handed” mode may be more effort-saving or convenient, it sacrifices crucial features (such as balance and realism), whereas switching to “full-body” movements, though offering greater immersion or realism, also raises operational difficulty, physical effort, and learning costs, which could deter users. Walking is the only movement participants prefer to perform with their entire body. They do not perceive walking as physically taxing, unlike jumping. As P6 mentioned, “Walking is a daily habit, and walking with the full body is the most intuitive and natural,” making full-body walking the easiest to learn and understand, fulfilling the desire for a more realistic virtual movement experience through bodily movement.

There seems to be a threshold to the level of effort that participants exert in virtual reality movements, which is higher than in video games. This is reflected in their preference for using both hands or full-body movements for many movements, but when the effort required is too high, they tend to prefer single-handed controllers or gestures. The choice of movement preference is also closely related to task complexity. If the task is simple enough, bare-handed, single-handed gestures become the optimal choice. Participants prefer single-handed controllers if the task involves complex processes or high precision. Finally, if the movement consists of grasping objects in real life, the haptic feedback offered by hand controllers becomes another reason for their choice.

6 Limitations and Future Works

This research broadens the applications of gesture-inspired studies and reevaluates the pros and cons of barehanded-based and controller-based methods from a novel perspective. However, some limitations remain:

- (1) As this is an exploratory study, the experiment involved only ten participants with prior virtual reality experience, which can lead to bias compared to the general population. Future research will verify the reliability of these conclusions on a larger scale.
- (2) The instructions were delivered to participants via voice commands, occasionally leading to ambiguities in interpretation, such as the term “wave,” resulting in two utterly different task movements. Besides this, in spellcasting, participants devised

different gestures depending on their imagined scenarios and specific implementations, which enhanced the diversity of spellcasting styles but did not support clear movement designs. In future studies, we will explore various instructional cues tailored to specific experimental goals and evaluate their impact on the final experimental results.

7 Conclusion

This study first constructs a set of 10 tasks in a virtual reality action consensus gesture set, including both bare-handed and controller-based operations, filling the gap in the virtual reality interaction gesture research regarding the combination of controllers and hand-free actions. Compared to previous virtual reality gesture studies, the action selection in this study overcomes the limitations of specific contexts or interaction actions by systematically incorporating controller operations, providing broad reference value. The study found that although most participants preferred bare-handed operations, controller interaction became more attractive in contexts involving haptic feedback, multi-functionality, high-precision tasks, and the need for reduced effort. Furthermore, participants were more willing to invest additional effort to perform actions in the virtual reality environment than in the Kinect gaming environment. This finding provides designers with a guide to methodologies and key considerations for future VR gesture design, offering critical practical insights.

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